

Brainstorming & Idea Prioritization

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. Problem Ideation

Commercial credit departments receive massive volumes of daily credit card applications. Processing manually introduces operational bottlenecks and high error rates.

Our idea focuses on utilizing modern predictive machine learning classifiers to score applications instantly, optimizing human underwriting time.

2. Candidate Models & Evaluation Criteria

- Logistic Regression: Fast baseline, low complexity.
- Decision Tree: Transparent, non-linear partitioning.
- Random Forest: Ensemble model, robust to outliers.
- XGBoost: Advanced gradient boosted trees for peak performance.

3. Selection Matrix

Models were prioritized based on accuracy, recall (minimizing default risk), and low execution latency. The training pipeline evaluates all models and selects the highest accuracy classifier automatically for production serving.